Ushtrime ND10b

2.1.9 *Në temperaturën 0^0C thupra e zinkut ka gjatësinë $l_0 = 500mm$, ndërsa e bakrit gjatësinë $l_{01} = 501mm$. Njehsoni në cilën temperaturë do të kenë gjatësira të njejta. Koeficienti i bymimit termik për zink ka vlerën $\alpha_1 = 1{,}06 \cdot 10^{-5} K^{-1}$, ndërsa e bakrit $\alpha_2 = 0{,}6 \cdot 10^{-5} K^{-1}$.*

Zgjidhje:

Të dhënat:

$t' = 0°C$

$l_0 = 500\ mm = 500 \cdot 10^{-3}\ m$ (Zn)

$l_{01} = 501\ mm = 501 \cdot 10^{-3}\ m$ (Cu)

$\alpha_1 = 1{,}06 \cdot 10^{-5} \frac{1}{K}$

$\alpha_2 = 0{,}6 \cdot 10^{-5} \frac{1}{K}$

$t = ?$

$l_1 = l_2$ kushti i dhënë

$$\left.\begin{aligned} l_1 &= l_0(1+\alpha_1 t) \\ l_2 &= l_{01}(1+\alpha_2 t) \end{aligned}\right\} \quad l_0(1+\alpha_1 t) = l_{01}(1+\alpha_2 t)$$

$$l_0 + l_0\alpha_1 t = l_{01} + l_{01}\alpha_2 t$$

$$l_0\alpha_1 t - l_{01}\alpha_2 t = l_{01} - l_0$$

$$(l_0\alpha_1 - l_{01}\alpha_2)\, t = l_{01} - l_0 \ /:(l_0\alpha_1 - l_{01}\alpha_2)$$

$$t = \frac{l_{01} - l_0}{l_0\alpha_1 - l_{01}\alpha_2} = \frac{(501-500)\cdot 10^{-3} m}{500 \cdot 1{,}06 \cdot 10^{-8} \frac{m}{K} - 501 \cdot 0{,}6 \cdot 10^{-8} \frac{m}{K}}$$

$$t = \frac{1 \cdot 10^{-3} m}{530 \cdot 10^{-8} \frac{m}{K} - 300{,}6 \cdot 10^{-8} \frac{m}{K}} = \frac{1 \cdot 10^{-3} m}{229{,}4 \cdot 10^{-8} \frac{m}{K}}$$

$$t = \frac{1}{229{,}4} \cdot 10^5 K = 4{,}359 \cdot 10^{-3} \cdot 10^5 K = 435{,}9 K$$

$T = 435{,}9 K$, $\boxed{l_1 = l_2}$

Të dhënat:

$l_0 = 1\ m$

$m = 1\ kg$

$\Delta l = 1\ mm = 1 \cdot 10^{-3} m$

$\alpha = 1{,}2 \cdot 10^{-5} \frac{1}{°C}$

$c = 460 \frac{J}{kg \cdot °C}$

$Q = ?$

$Q = C \cdot \Delta t$

$C = c \cdot m$

$\boxed{Q = c \cdot m \Delta t}$

$l = l_0(1 + \alpha \Delta t)$

$l = l_0 + l_0\alpha \cdot \Delta t$

$l - l_0 = l_0\alpha \cdot \Delta t$

Δl

$\Delta l = l_0\alpha \cdot \Delta t \ / : l_0\alpha$

$$\Delta t = \frac{\Delta l}{l_0 \alpha} = \frac{1 \cdot 10^{-3} m}{1 m \cdot 1{,}2 \cdot 10^{-5} \frac{1}{°C}} = \frac{1}{1{,}2} \cdot 10^{-3} \cdot 10^5 °C$$

$\boxed{\Delta t = 83{,}33°C}$

$$Q = mc \cdot \frac{\Delta l}{l_0 \alpha} = 1 kg \cdot 460 \frac{J}{kg °C} \cdot 83{,}33 °C$$

$Q = 38331{,}8\ J = 38{,}331\ kJ$